Continuous Probability Distributions

GENERAL ASSEMBLY

Agenda

- Review of random variables and discrete distributions
- Continuous distribution rapid fire
- Continuous distributions in Python

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Recap

Earlier...

We studied **discretely distributed random variables**.



In your own words, what is a **random variable**?

Earlier...

We studied **discretely distributed random variables**.



In your own words, what is a **distribution**?

Rapid Fire: What are each of these?

- Uniform distribution
- Bernoulli distribution
- Binomial distribution
- Poisson distribution
- Brownie points: Any others?

The Differences

Density Plots

For discrete variables, we had a **probability mass function (pmf)** that said “at a given value, what is its probability occurring?”

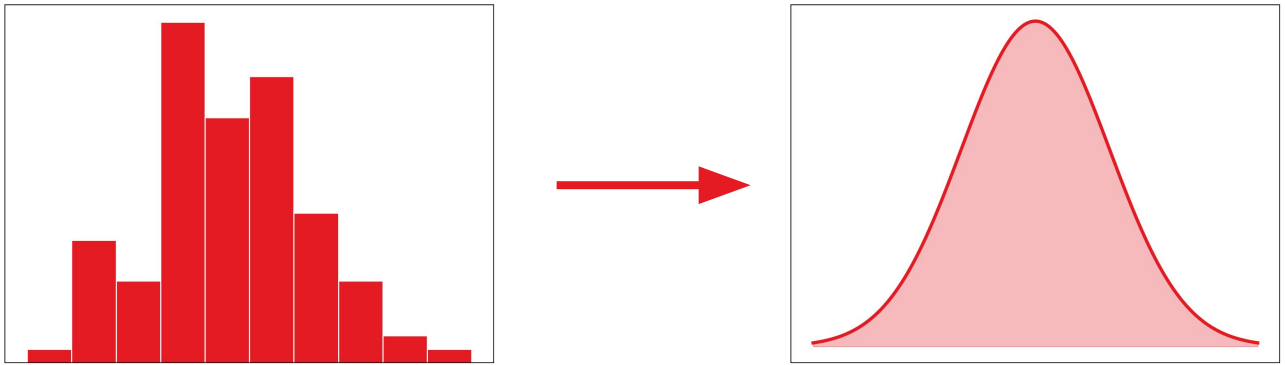
We don’t have such a thing for continuous variables.

Why?

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Density Plots

Instead of a pmf, we have a **probability density function (pdf)**. This is a function that describes the comparative probability of an event occuring. Since we can't find probabilities at a given point, **we find probabilities along intervals**.



Calculus Aside

If you've taken a calculus course before, this might sound like an **integral** to you. Surprise: it is!

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

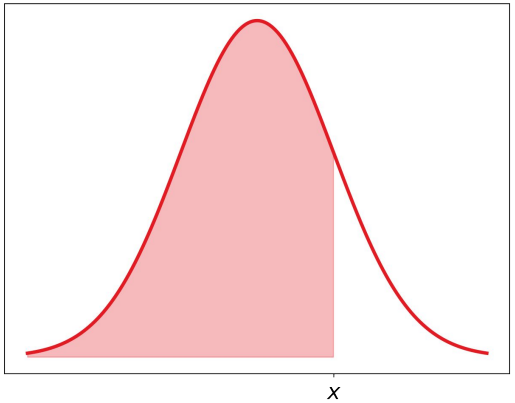
Of course, this is ***not*** a calculus class, so you won't be doing this by hand. Lucky you!

CDFs

Because of the calculus trickery, a lot of continuous probability is told via **cumulative density functions (CDFs)**. Simply put, a CDF of a distribution is the probability of getting at or below a certain value. Mathematically:

$$F(x) = P(X \leq x)$$

Pretty much all the “named” distributions have their pdfs and CDFs made for us in the SciPy library.

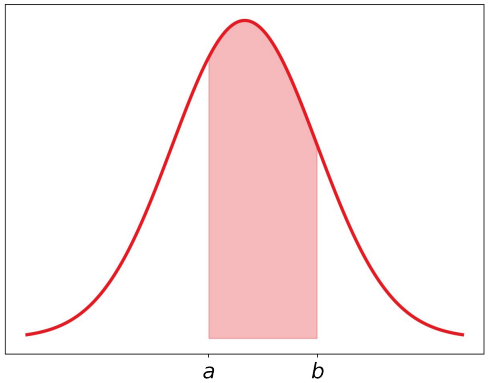


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Probabilities via CDFs

This is typically how we compute probabilities of continuous functions. There's a lot on this slide, so let's just soak it in:

$$P(a \leq X \leq b) = F(b) - F(a) = \int_a^b f(x)dx$$



The Common Continuous Distributions

The Common Continuous Distributions

On the next several slides, we'll be talking about some of the more major, well-known continuous distributions. This isn't all of them, of course. These are just the ones you'll see the most, and you'll see them all again later:

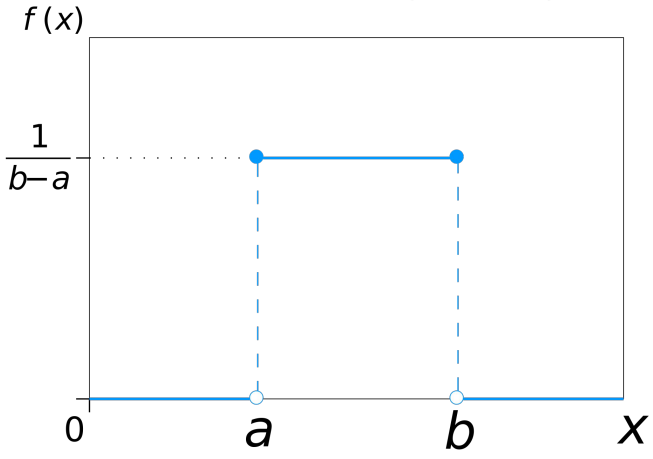
- Continuous Uniform
- Normal (and the Standard Normal)
- t
- Exponential
- Gamma
- Beta

The Continuous Uniform

If all numbers in a *continuous* interval [*a*, *b*] are equally likely, you could use a **continuous uniform distribution** to model this scenario.

pdf	$f(x) = \frac{1}{b-a}$
support	$a \leq x \leq b$
mean	$\frac{a+b}{2}$
variance	$\frac{(b-a)^2}{12}$

$$X \sim U(a, b)$$



Stolen unapologetically from Wikipedia.

The Normal Distribution

Maybe the most well-known is the **normal distribution** (sometimes known as the **Gaussian distribution**). It is bell-curved, symmetric, and entirely described by its mean and variance μ and σ^2.

We typically **assume** something is normally distributed when we have a continuous number that varies from some mean, equally likely above and below.

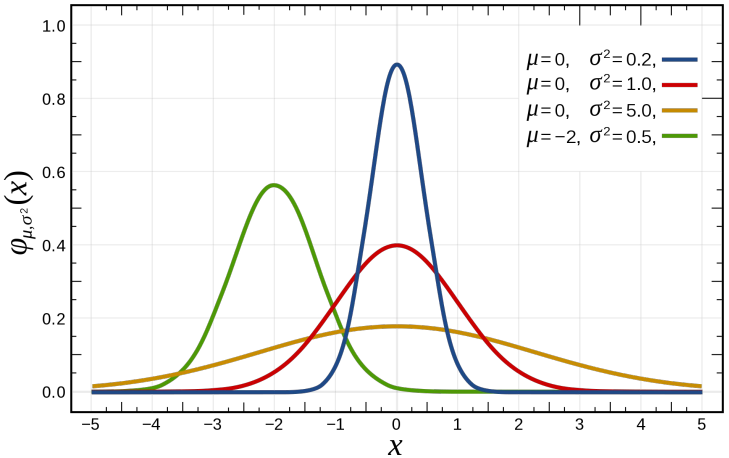
While it's actually surprisingly **rare** for something to *really* be normally distributed, it makes a very good approximation in a lot of real scenarios. It's also a very fundamental distribution to a lot of statistical theory.

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The Normal Distribution

pdf	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}$
support	$-\infty < x < \infty$
mean	μ
variance	σ^2

$$X \sim N(\mu, \sigma^2)$$



Stolen unapologetically from Wikipedia.

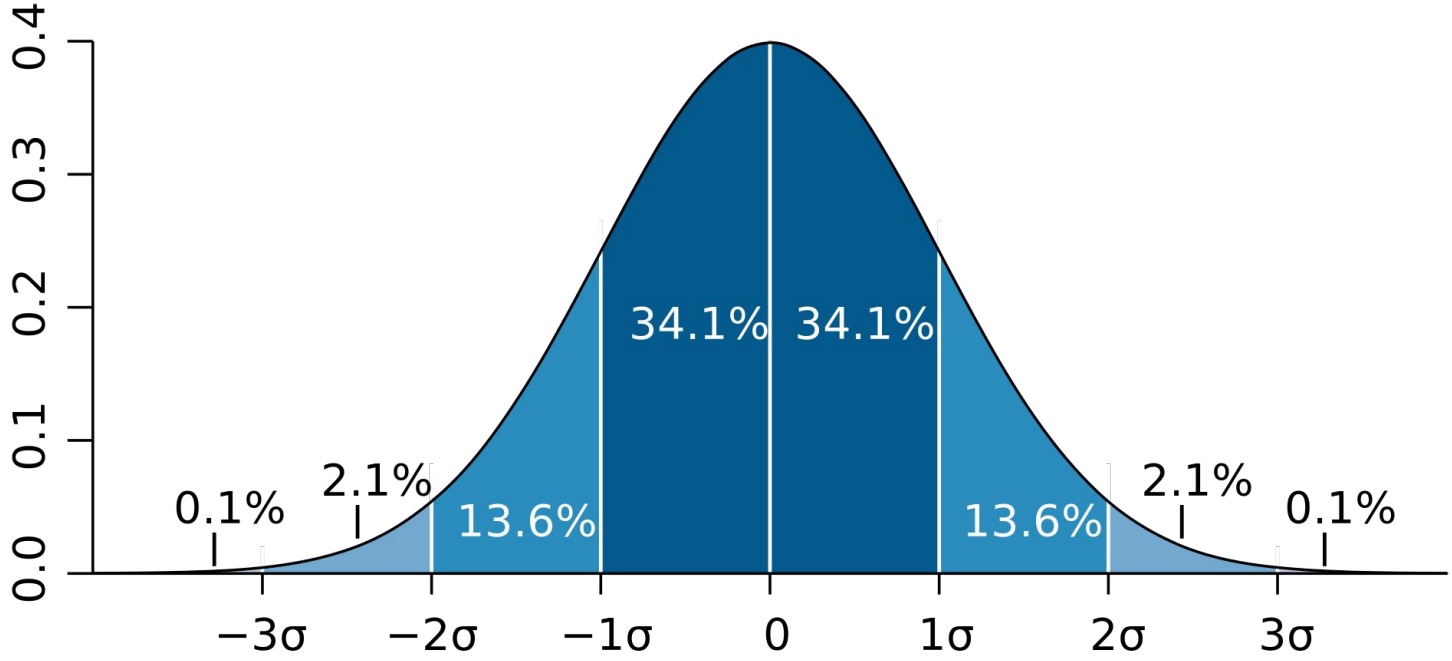
The Empirical Rule

The **Empirical Rule** (or the **68-95-99.7 Rule**) is a commonly used guideline for thinking about probabilities. It states the following about a **normal distribution:**

- The probability of an event occurring within 1 standard deviation from the mean (ie, $-1 \leq z \leq 1$) is **68%**.
- The probability of an event occurring within 2 standard deviation from the mean (ie, $-2 \leq z \leq 2$) is **95%**.
- The probability of an event occurring within 3 standard deviation from the mean (ie, $-3 \leq z \leq 3$) is **99.7%**.

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The Empirical Rule, Visualized

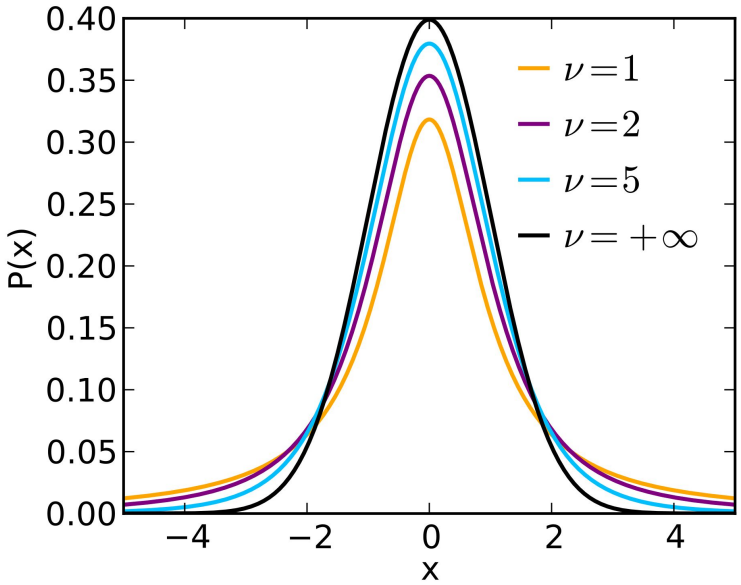


Stolen unapologetically from Wikipedia.

Student's *t*-distribution

This is a weird one. Graphically, it looks a *lot* like the standard normal.

However, it has an additional parameter: its **degrees of freedom**, usually denoted *df* but sometimes as the Greek letter ν.



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Student's *t*-distribution

In fact, as the number of degrees of freedom increases, it approaches the standard normal asymptotically. That is,

$$\lim_{\nu \to \infty} T_\nu = Z$$

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Student's *t*-distribution

What's this for? Inference!

Soon, we're going to encounter scenarios where we want to **find** the unknown true population μ, and we're going to estimate it by taking samples. However, in order to use the normal distribution, we'd need to know the true population SD, σ. That's kinda nonsense, right? When would we ever know σ but not μ?

The *t*-distribution allows us to use the sample SD instead of the population SD to do this sort of work. Keep it in mind!

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Aside: William Sealy Gosset (aka Student)





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The Exponential Distribution

The **exponential distribution** is a heavily right-skew distribution. It is very commonly used to model **waiting-time** data.

It has only one parameter, λ... or is it $\square$?

The Exponential Distribution

Suppose Tim gets a spam email every 10 hours. You could model the time between spam emails via the exponential distribution with *rate* 1/10. Many resources instead prefer to just use this 10 as the parameter, since it also happens to be the mean. That is...

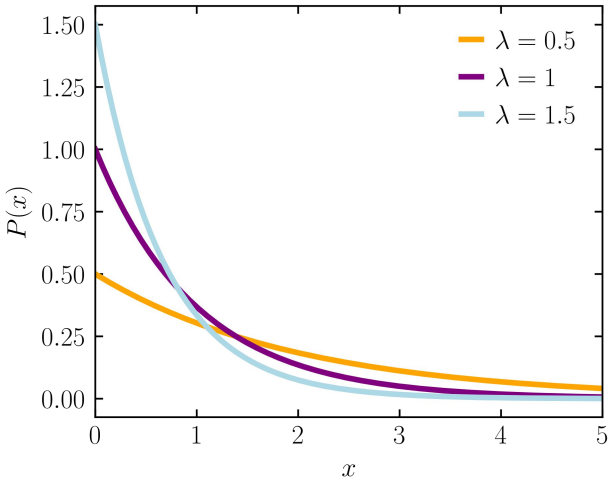
$$X \sim \text{Exp}\left(\frac{1}{10}\right) \quad \text{and} \quad X \sim \text{Exp}(10)$$

Are both commonly written to mean the same thing. Be mindful of the context!

The Exponential Distribution (mean notation)

pdf	$f(x) = \frac{1}{\beta} e^{-x/\beta}$
support	$x > 0$
mean	β
variance	β^2

$$X \sim \text{Exp}(\beta)$$

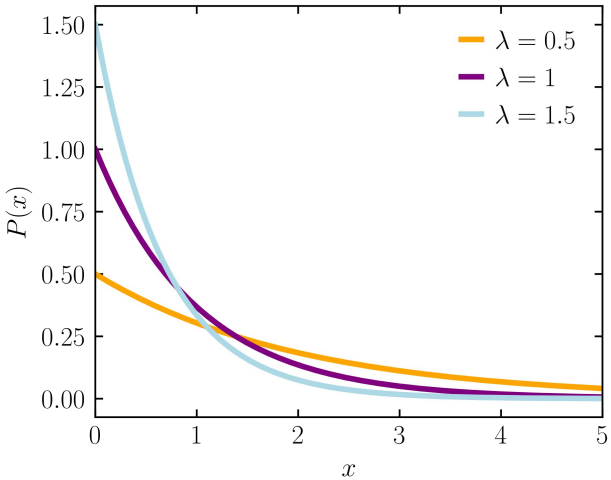


Stolen unapologetically from Wikipedia.

The Exponential Distribution (rate notation)

pdf	$f(x) = \lambda e^{-\lambda x}$
support	$x > 0$
mean	$\frac{1}{\lambda}$
variance	$\frac{1}{\lambda^2}$

$$X \sim \text{Exp}(\lambda)$$



Stolen unapologetically from Wikipedia.

Exponential and Poisson

The exponential distribution and the Poisson distribution have an interesting relationship.

- Poisson can model **how many times an event occurs in a fixed timeframe**
- Exponential can model **how long it takes for an event to occur**
  ➜ So, the time in between Poisson events is exponentially distributed!

Modeling these things in conjunction is called a **Poisson process**. It's a topic of study in **stochastic statistics**, which is way beyond our scope. But it's cool!

The Gamma Distribution

I won't spend a lot of time on this. It's very similar to the exponential distribution except with an additional **shape** parameter, α. I'll include only two interesting gamma facts that are worth knowing at this time:

- Just like exponential, it has support x > 0, is skew right, and is used to model waiting times.
- ... Actually, it's used to model α consecutive waiting times! That is... if we have α exponentials:

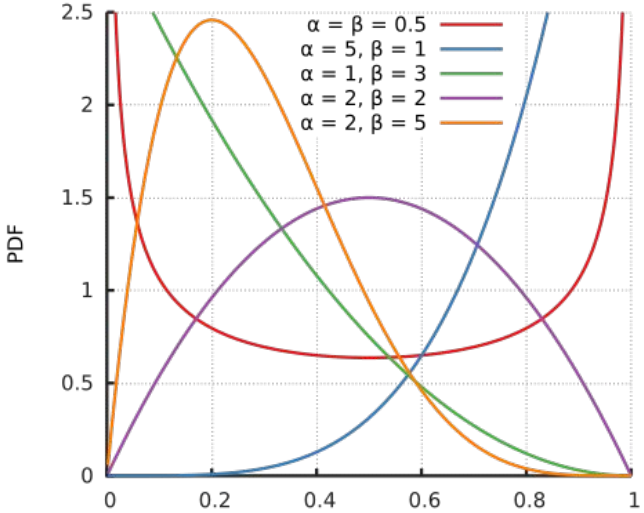
$$X_1 + \cdots + X_\alpha \sim \text{Gamma}(\alpha, \beta)$$

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The Beta Distribution

We won't see this too often, but good to know it exists.

- Two parameters α and □. In practice they're often the same number. They control how "bowl" or "hill" shaped the pdf is.
- Since the support is continuous between 0 and 1, the Beta distribution is often used to model probabilities in **Bayesian statistics**, which we'll get to much later.
- The standard uniform is a special case: Beta(1, 1) = U(0, 1)



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Finale

RV Relationships

You may have noticed there's a lot of relationships between various random variables. You have touched only the tip of the iceberg, my friend.

Check this out.

Bonus: Multivariate Distributions

Statistics is a wide-open field, and we're really only learning what's prerequisite for the major data science topics in our course. There are also **multivariate distributions** that are represented by vectors instead of scalars. I'll briefly mention two interesting ones:

- The **multinomial distribution**: What if a trial had more than two options?
- The **multivariate normal distribution**: What if you're generating a vector of values from a normal distribution... but they're not independent?

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